

# Karthikeya Reddy Basavanagoudgari

(929) 789-1169 | [bkarthikeyareddy22@gmail.com](mailto:bkarthikeyareddy22@gmail.com) | [LinkedIn](#) | [GitHub](#)

## EDUCATION

### The City College of New York (CUNY)

New York, NY

*Bachelor of Science in Computer Science* GPA: 3.3

*Expected May 2027*

- **Relevant Coursework:** Data Structures, Algorithms, Software Engineering, Operating Systems, Probability & Statistics, Linear Algebra

## PROJECTS

### Benchmarking MARL in Augmented Reality | *Python, PettingZoo, PPO/MADDPG*

May 2026

- Designed a multi-agent alternative to single-item AR previews (e.g. Amazon Room Decorator), collaborating with a 2-person team to build a simulator with 3 coordinated agents (layout, style, budget), achieving 50–55.6% space utilization across 5 room types against a 50% target
- Merged a 12,600+ image IKEA dataset with catalog pricing data into a unified spatial-cost dataset for a 15x15 grid simulation
- Configured PPO training (Stable-Baselines, 100,000 timesteps) on a vectorized multi-agent environment as a baseline

### BudgetBuddy | *Next.js, React, Supabase, Jest, GitHub Actions*

Dec 2025

- Collaborated with a 5-person team to build and deploy a full-stack budget tracking app, implementing 15 REST API routes across expense, income, and budget management
- Wrote 816 lines of automated tests (Jest) across 4 suites covering auth, expenses, income, and budget logic, integrated into 2 GitHub Actions CI/CD pipelines for testing and database migrations
- Followed a PR-based code review workflow with branch protection and squash merging, resolving issues and reviewing teammates' code across 3 iterative schema migrations

### Tutorly | *React, TypeScript, Supabase*

Apr 2025

- Built an 8-page full-stack peer tutoring marketplace with dual-role (student/tutor) authentication, backed by a 7-table Supabase schema covering bookings, availability, messaging, and reviews
- Built real-time messaging (Supabase Realtime channels) and a booking/availability system with status tracking
- Built dynamic tutor search and filtering by subject and university, surfaced through a dedicated ratings and reviews system

### GoldLinks | *React, Next.js, Flask, MySQL, OpenAI API*

Nov 2024

- Collaborated with a 6-person team to build a dual-role (customer/business-owner) platform for real-time gold price and store comparison, integrating an OpenAI-powered chatbot to help customers learn about gold types and products
- Architected a normalized (2NF) MySQL schema of 8 interconnected tables and wrote 8 stored procedures for automated dummy-data seeding, covering inventory, pricing history, and subscriptions
- Integrated React/Next.js with a Flask backend, incorporating OAuth authentication and cloud deployment

## RESEARCH

### Co-Author, Prof. Bandyopadhyay's AI Agents Lab

New York, NY

*The City College of New York*

*Apr 2026*

- Listed as co-author on early-stage lab projects in AI-driven protein binder design (BindCraft, AlphaFold2-based pipeline) and mathematical discovery; building technical foundation through lab meetings and literature review

## LEADERSHIP & INVOLVEMENT

### Fellow, CUNY Tech Prep

New York, NY

Competitive full-stack web development training program covering modern JavaScript and agile engineering. *Jun 2026 – Present*

### Social Media Chair, Outdoors Club

New York, NY

*The City College of New York*

*Aug 2024 – Aug 2026*

- Manage digital content strategy and communications for club events, increasing club visibility

### Event Volunteer, NY AI Agents Summit

New York, NY

*CCNY & CUNY ASRC*

*Apr 2026*

- Supported speaker coordination for a two-day summit featuring industry leaders from Google, NVIDIA, Microsoft Research, DARPA, and NASA

## TECHNICAL SKILLS

**Languages:** C++, Python, SQL, JavaScript, TypeScript

**Backend/Data:** Pandas, NumPy, Flask, Supabase, OpenAI API

**Frontend:** React, Next.js, HTML, CSS

**Databases:** MySQL, PostgreSQL (Supabase)

**Tools:** Git, GitHub Actions, Jest, VS Code, Excel